Make the delivery

Name: ______________________

Date: ______________ **Period:** ______________

Today I can...

Join your tested parts and check the whole delivery three times.

Words to know

Success criteria: The things that must happen for a test to count as a success.

Version: One saved design or program before you change it.

Reliable: Works as expected again and again.

LOOK AT THE IDEA

Gather your materials

- Checked base and the selected holder from lesson 3
- The same light paper parcel
- White lane and red strip from lesson 6
- Removable tape, ruler and saved code

Build the complete delivery course

Build with the power off. Tick each step when it is done.

1. Turn the robot off. Attach the tested holder to the approved mounting points. Put the paper parcel low in the center. Check that no part touches a wheel or covers the sensor.
2. Keep the white approach and red strip from lesson 6. Tape a delivery rectangle around the stopping area, large enough for the whole robot.
3. Keep the lane clear. For this test, use a short straight approach to the red strip. Add a longer route only after this works.
4. Use the checked color-stop program from lesson 6. Keep its short moves, Stop steps and maximum of 20 checks.

Sketch the setup. Label the parts you will check.

Finish the setup

5. Write three rules before driving: parcel stays on; robot stops inside the delivery space; robot hits nothing. A run passes only when all three happen.
6. Run the delivery three times from the same start. Write Yes or No for each rule after each run. Keep the failed runs in the notes.
7. Choose one change, such as a lower holder wall or wider red strip. Keep the other parts and code the same. Try another set of three runs.
8. Save the best-tested setup. Keep the base ready for the final build choice.

- [ ] Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Use the complete lesson 6 color-stop program.

2. White permits one 0.1-second slow move and then Stop. Red or any other reading ends the test.

3. Stop and end after at most 20 checks. Never restart within the same test.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

keepGoing is a value the program remembers: Yes allows another check; No keeps the robot stopped.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

Test it fairly

Name: ______________________________

Date: ____________________ **Period:** ____________________

Change one thing between versions. Keep everything the same during each set of three runs.

A successful test

- ☐ At least two of three runs meet all three rules.
- ☐ The robot still passes the red-and-unknown stop checks.
- ☐ The team can name one change and the result it produced.

Record every result

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

Which result stands out?

____

Change one thing. Try again.

The one thing we changed:

We kept these things the same:

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

What improved when you changed one part of the delivery?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.